Multidimensional Goodhart

Measurement Dimensions, Exchange Rates, and Hidden Harm

Contents

1 The Measurement-Dimension Question

1.1 More metrics, fewer metrics, different metrics

Every scorecard owner eventually faces the same design question. A measure is under pressure and the gaming has begun. Should the scorecard measure more things, fewer things, or different things — and will the hidden damage from gaming shrink, grow, or just move?

Two folk intuitions answer immediately, in opposite directions. The first says more metrics leave nowhere to hide: every added dimension closes an escape route, so a gamed system should be measured more finely. The second says every metric is a new attack surface: every added dimension is one more channel that can be manufactured, so measurement should stay sparse and robust. Both sound like engineering judgment. They cannot both be right as stated, and neither is.

In the one closed model where this book can answer the design question completely, the outcome is decided not by how many dimensions are measured but by each gaming channel's exchange rate: how much hidden harm the channel does per point of score it produces. Changing the measured set conserves hidden harm exactly when every active channel does the same hidden harm per score point. When the rates differ, adding or removing metrics reduces, increases, or reroutes harm according to which channels the design leaves cheap. The dimension count never enters by itself. That statement is made precise as T5 once the model behind it has been declared. Everything before the theorem exists to establish what such a declaration requires; everything after it, to say when the calculation does and does not transfer.

1.2 The warning and what it does not identify

Goodhart's law is usually introduced through the compressed formulation "When a measure becomes a target, it ceases to be a good measure" [23]. The original macroeconomic warning is sharper about policy use: "Any observed statistical regularity will tend to collapse once pressure is placed upon it for control purposes" [8]. Both describe a real empirical phenomenon: once a measure is used for control, the residual error between proxy and target can acquire direction, support, active constraints, exchange rates, tail behavior, and response dynamics. But none of those shapes is identified by the claim that the proxy became a target — and neither is the answer to the design question above. The warning says a controlled measure will stop tracking its target. It does not say what an observed score movement means, and it does not say whether measuring differently would help.

1.3 Five stories, one score path

The identification problem is concrete. A district announces that school funding will depend on a test-score metric. Next year the average score rises. That one fact is compatible with several different stories:

- stronger schools entered the funded pool;

- the same schools taught the underlying material better;
- the same schools drilled the test format while learning did not improve;
- low-scoring students were excluded from the tested population;
- reporting errors were repaired with little hidden harm.

The same score path fits all five stories; they differ in who changed and through which channel. Two channel families matter most and recur through the book. **Selection** changes who is represented: the pool is re-ranked or re-weighted while each unit keeps behaving as before. **Intervention** changes what a fixed unit does: the same school alters its teaching, its test preparation, or its reporting. Selection and intervention can produce identical marginal score movement, and they do not license the same mathematics. Hidden harm is no more readable off the path than mechanism is: the format-drilling story and the better-teaching story can move the score by the same amount.

So observed score movement underidentifies both mechanism and harm, and the designer's question cannot be answered by watching the score. The book's program follows: declare the response model — who can respond, through which channels, at what cost, with what hidden harm — and then calculate. There is no shortcut around the declaration; the next part shows that every unconditional law we tried to prove in its place is false. The part after that states the declaration itself, and then the design question gets its answer.

2 No Generic Law

The failed stronger claims determine the framework's shape: they rule out a single n-dimensional Goodhart law and leave conditional calculations tied to declared response models. Each failure below also dictates a field of the declaration introduced in the next part. The failures come in two tiers: stronger laws refuted by explicit counterexample, and methodological boundaries — claims that fail because marginal data cannot identify what they assert. The load-bearing members of both tiers follow in paired form; the full gallery is tabulated in the appendix.

2.1 Refuted by counterexample

Not: hidden harm scales with the number of unmeasured dimensions. Hidden harm does not scale with $\dim(\ker \varphi)$ by itself. In a pure selection model, if hidden coordinates are independent of the selected proxy, thresholding does not move them — however many of them there are. What survives is a coupling question: unmeasured dimensions matter through their dependence on the selected proxy and through the declared value metric, not through their count.

Not: more measured dimensions means more error — or less. Additive and conjunctive aggregation can give opposite comparative statics from the same primitive action channels. What survives is an aggregation-conditional calculation: the comparative static has a sign only after the aggregation rule is declared.

Not: signed aggregate hidden error measures damage. Positive and negative hidden movements can cancel in a signed aggregate. What survives is a declaration requirement: a value functional, norm, tail risk, or domain loss must be named before "damage" has a magnitude.

Not: baseline covariance predicts response under pressure. With $P = Z$ and $H = Z^2 - 1$, baseline covariance between proxy and hidden variable is zero, yet threshold or Boltzmann selection moves H. The example shows why zero baseline covariance is not enough at finite pressure. What survives is covariance as a local velocity: the derivative at zero pressure, with finite-pressure behavior depending on the whole tilted path.

Not: what an agent can afford to game is what gaming costs the world. A convex score-deficit cost bounds the proxy movement a unit can privately afford; it says nothing about hidden harm. Two channels with equal private cost and equal score weight can carry arbitrarily different harm. What survives is a division of labor: affordability calculations license affordability claims, and welfare claims need a declared harm functional.

2.2 Methodological boundaries

Not: the data says whether the response was selection or intervention. The selection/intervention split is not identifiable from marginal score movement, and it is relative to the declared type representation: enrich the types enough and every intervention looks like selection over richer types. What survives is the split as a declared, defended field of the claim, supported by repeated-type, exposure, action-trace, or structural evidence.

Not: optimization pressure drives failure toward simple shapes. Minimum-complexity attraction is not generic: selection follows baseline tails, and intervention follows costs, caps, fixed charges, and affordances. What survives is the response-shape question as a conditional one, taken up with the supporting calculations.

Two further boundary entries — absolute continuity as an intervention marker, and the coordinate-dependence of the selection bound — are in the appendix table; they refine the calculations rather than redirect the design question. Together the tiers explain the shape of what follows: no unconditional law survives, so the route to the exchange-rate answer runs through a declaration discipline.

3 The Response-Modeling Contract

3.1 What the contract does

The school-score example left five stories on the table for one score path. The response-modeling contract is the instrument that separates them: a declaration, made before any calculation is imported, of who can respond, what they can do, at what cost, and what hidden value is at stake. The contract has three plain-language jobs:

1. State what the claim wants to conclude.
2. State what response story would make that conclusion meaningful.
3. State what evidence would distinguish that story from nearby alternatives.

Different stories require different mathematics. Selection bounds apply to reweighting a fixed population. Intervention bounds need actions, costs, and stakes. Welfare claims need a hidden value or harm model. The contract is the rule that says which story is being claimed before the corresponding calculation is imported.

One status distinction does most of the work: nearly every contract field is **declared or estimated, not observed**. “Hidden” in this book means hidden from the scorecard, not necessarily from the analyst. Harm rates, value weights, response channels, and action costs are inputs the analyst supplies and defends — from audits, side data, or structural knowledge — and every licensed calculation is conditional on those declarations. The theorems convert declared primitives into conclusions; they do not extract the primitives from the score path.

The contract is not a theorem about what proxy pressure will do. It is a methodological definition: the minimum declaration needed before a proposed Goodhart calculation can be evaluated. Its content comes from the failures of the previous part — marginal score movement does not identify hidden welfare, the type weights W_θ, the response kernels K_θ, the action costs, or the aggregation rule — so the declaration has to supply what the data cannot.

3.2 The notation

Let S be a state space. A target map $G : S \to \mathbb{R}^m$ records target-relevant state and a proxy map $P : S \to \mathbb{R}^k$ records measured features. The intended correspondence is $\varphi : \mathbb{R}^m \to \mathbb{R}^k$, with residual $\varepsilon(s) = P(s) - \varphi(G(s))$.

Two gaps matter. The **dimension gap** is the part of target variation the proxy map cannot see: when φ is linear it is carried by $\ker \varphi$, and in general by variation within the level sets of φ. The **observation gap** is residual measurement artifact inside the measured domain, represented by ε. A scalar score may hide both gaps at once.

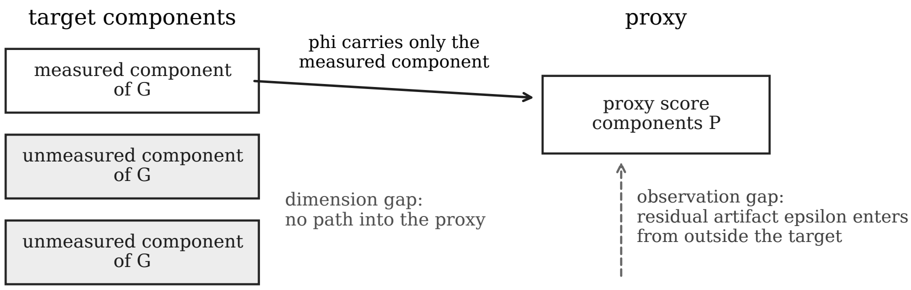

Figure 1: Dimension and observation gaps are different contract fields. The schematic fixes the vocabulary; it draws no quantitative conclusion.

A type space U is the model's description of the units before the policy response. In the school example, $u \in U$ might be one school with baseline traits such as size, neighborhood, student mix, prior resources, and administrative capacity. It contains the attributes treated as fixed for the comparison; choices made because of the policy are not part of u. The baseline type law ν says how common those school types are. The observed state $s \in S$ contains what the evaluation later sees, such as test scores, curriculum, student participation, and hidden learning outcomes when they are measured.

The response kernel $K_{\theta(ds \mid u)}$ is the conditional law of the observed state for a fixed type u after policy exposure θ — read the conditioning in that order: fix the type, expose it to the policy, and the kernel returns the distribution over observed states. It is where the model puts what the same school does after seeing the funding rule. The selection weight $W_{\theta(u)} \geq 0$ is where the model puts changes in which fixed types appear in the realized population. With these objects, the induced law is

$$\mu_{\theta(A)} = \frac{\int W_{\theta(u)} K_{\theta(A \mid u)} \nu(du)}{\int W_{\theta(u)} \nu(du)}.$$

The formula averages fixed-type response laws over the baseline type law, weighted by participation, and normalizes. Pure selection means $K_\theta = K_0$ for ν-almost every type and policy dependence enters only through W_θ. Intervention means K_θ changes on a positive-ν set of fixed types. Mixtures are allowed, but they must be named.

3.3 The declaration

A Goodhart claim must therefore declare the primitives that make it a claim rather than a retrospective label:

Contract.

- **Claimed output:** What are you trying to conclude: score drift, hidden drift, a distribution, a selection/intervention label, an intervention feasibility bound, a welfare comparison, or a response-shape prediction?
- **Type representation:** What is one fixed unit $u \in U$? For a school score, is u a school before the funding rule, a student, a classroom, or a district? Which attributes are fixed, and which are later choices?

- **Baseline behavior:** What does a fixed type produce before the policy response? Declare ν and $K_0(ds \mid u)$: the baseline distribution of observed states conditional on type.
- **Policy exposure:** What creates pressure, and who sees it: threshold, ranking, scorecard, prize, penalty, benchmark, or feedback signal indexed by θ?
- **Response channel:** Does θ only reweight fixed types through $W_{\theta(u)}$, change fixed-type behavior through $K_{\theta(ds \mid u)}$, or both? Selection changes who is represented; intervention changes what a fixed unit does or produces.
- **Action/search geometry:** If fixed-type behavior changes, what actions are available, what do they cost, what caps or fixed charges exist, what search process is used, and what stakes V make response worthwhile?
- **Proxy/target relation:** What proxy P is optimized, what target G or hidden quantity H is protected, what relation $P \approx \varphi(G)$ is intended, and where are the dimension and observation gaps?
- **Aggregation:** If there are multiple proxy components, how are they combined: additive weights, thresholds, conjunctive gates, Pareto rules, lexicographic rules, or an institution-specific formula?
- **Hidden value or harm:** What makes a response good or bad beyond the proxy: hidden coordinates, scalar value vector, norm, loss, or harm rates h_j?
- **Evidence standard and falsifier:** What observations would distinguish the claimed story from nearby alternatives, and what observation would make this contract the wrong one?

Each primitive is in the list because some claimed output needs it. Hidden value or harm is needed for any welfare conclusion. The selection weight W_θ is needed for selection claims, and the response kernel K_θ for fixed-type response claims. Action costs and stakes are needed for intervention feasibility. The aggregation rule is needed before any scorecard comparison has content. A claim that skips a primitive is implicitly asserting that its output does not depend on it.

The first row is load-bearing. The contract is not a single formula with one fixed output. It is an adequacy test for a proposed output. A scalar hidden drift claim needs less information than a full response kernel. A distribution over k discrete states has $k-1$ degrees of freedom. A hidden drift vector in $\mathbb{R}^m$ needs m value-relevant coordinates. A response-kernel claim can be infinite-dimensional unless the model restricts it. A selection/intervention label has a small output alphabet, but it is causal: it cannot usually be read from the marginal score path.

The type representation is part of the empirical claim, not notation to hide inside the model. If U includes each school's whole future response plan, it can make every intervention look like selection over richer types. If U is too coarse, stable heterogeneity can look like a kernel change. The contract therefore has to defend why the chosen u is fixed for the comparison and why omitted variation belongs in K_θ, W_θ, or the action model. The induced marginal law μ_θ usually cannot distinguish those choices by itself.

3.4 Contract adequacy

After the output is named, the contract should pass a small information accounting check:

Contract adequacy.

1. State the output object and its degrees of freedom or identifying data.
2. List the raw primitives the claim uses: type law, kernels, weights, proxy and target maps, aggregation, action costs, stakes, hidden value, and evidence.
3. Subtract constraints, such as probability normalization, declared equality $K_\theta = K_0$, additive aggregation, convex cost, or fixed deficit.
4. Subtract redundancies, such as the arbitrary scale of W_θ or coordinate rescalings that leave the represented situation unchanged.
5. Check units: costs must be comparable to stakes, proxy gains to score deficits, and hidden harm rates to the declared value units.
6. Check rank or relevance: independent changes in the declared inputs should affect the claimed output, unless the redundancy has been stated.
7. Check invariance: the conclusion should not depend on arbitrary naming, coordinate splitting, units, or post-hoc enrichment/coarsening of U.

In shorthand, a contract is adequate only if

$$\text{true input information} = \text{raw variables} - \text{constraints} - \text{redundancies}$$

is enough to identify the claimed output up to the intended invariances. If two meaningfully different response stories still satisfy the same declared inputs, the contract has not licensed that output. It may still license a weaker output: for example, an aggregate score path may license a monitoring trigger while not licensing a welfare verdict or a selection/intervention classification. The school-score path is exactly such a case: the declared inputs — a score rise and a known funding rule — leave all five response stories standing, so the licensed output is “investigate,” not a verdict on any one story.

The contract is demanding by design. It prevents the framework from inferring welfare, hidden target movement, or a response channel from marginal score movement alone.

4 The Exchange-Rate Diagnostic

The design question of Part 1 — more metrics, fewer metrics, different metrics — has a complete answer in one closed model. This part declares the model, proves the answer, and then marks the boundary of the model's assumptions. The contract fields it consumes are the ones the previous part introduced: a fixed-type intervention channel, an action-cost geometry, an aggregation rule, and declared hidden harm rates.

4.1 The additive fixed-deficit model

The scorecard calculation needs more than "more metrics." It needs an additive score over a measured channel set M with weights w_j, separable quadratic action costs $\frac{a_j^2}{2\kappa_j}$, a fixed score deficit d that the gaming unit must close, and linear hidden harm $\sum_{j \in M} h_j a_j$. Each declared rate h_j, set against the score weight w_j, is the channel's exchange rate: how much hidden harm channel j does per unit of score it produces.

> **T5. Additive exchange-rate iff.** The fixed-deficit per-agent hidden harm of the cost-minimizing action is
>
> $$H_{M(d)} = d \frac{\sum_{j \in M} h_j \kappa_j w_j}{\sum_{j \in M} \kappa_j w_j^2}.$$
>
> Fixed-deficit harm is conserved across compared active measured sets exactly when $h_j = c w_j$ on those active channels.

The formula is a weighted harm-per-score average. The cost-minimizing unit closes the deficit by loading each channel in proportion to $\kappa_j w_j$ — the Lagrange condition prices every channel's score gain at equal marginal private cost — so each channel's harm rate enters weighted by exactly that load. The sums do not cancel in general because κ_j, w_j, and h_j vary by channel. The iff condition reads off the average: fixed-deficit harm is conserved across compared active measured sets exactly when every active channel does the same hidden harm per score unit, $h_j / w_j = c$.

That is the answer to the design question. Adding or removing measured channels conserves, reduces, increases, or reroutes the harm of cost-minimizing gaming strictly through the exchange rates of the channels the change makes cheap or expensive. The dimension count never enters by itself. This is the main additive-scorecard result, and it converts the folk intuitions of Part 1 into a checkable condition: before saying whether a measured-set change helped, hurt, or only re-routed harm, the hidden harm per score unit of the affected channels has to be declared or estimated. Holmstrom–Milgrom multitasking is the closest economics precedent [12] — an analogue for the aggregation-and-effort structure, not grounding for the theorem — and the contract here makes the hidden-harm exchange rate explicit.

4.2 Affordability is not the diagnostic

The diagnostic sits on top of an affordability calculation and must not be confused with it. The convex score-deficit budget (T4, stated with the supporting calculations) answers "can the unit afford the proxy movement?" under a declared private-cost model; the exchange-rate diagnostic answers "what does that movement do to hidden harm?" The two come apart cleanly: if two proxy channels have equal private cost and equal score weight but harm rates $(h, 0)$, cost minimization splits effort between them while hidden harm grows with h. Private cost is unchanged; value-weighted harm is not. Affordability is not welfare.

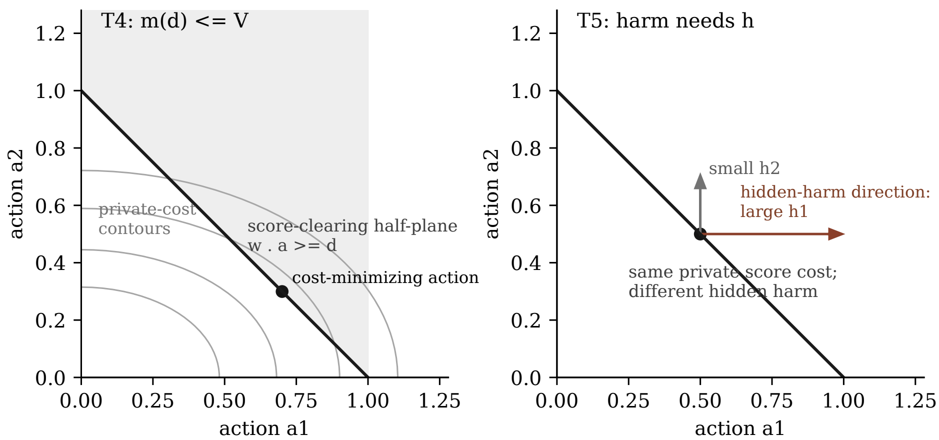

Figure 2: Left (T4): private-cost contours, the score-clearing half-plane $w \cdot a \geq d$, and the cost-minimizing action on its boundary. Right (T5): from that action, channels with equal private cost carry different hidden-harm directions, so reading the action as welfare movement requires the declared exchange rates h_j.

4.3 The non-transfer boundary

Each assumption in the model carries load, and the diagnostic does not survive losing any of them.

Additive aggregation. The weighted-average form exists because score contributions add. Conjunctive aggregation — requiring every measured component to clear a bar — is a different rule with different comparative statics, not a corollary; under it, harm can grow with the number of components.

Separable quadratic costs. Proportional loading, and with it the closed form of $H_{M(d)}$, is quadratic-cost algebra. Other convex costs still give an affordability budget but not the exchange-rate formula.

Fixed deficit, per agent. The result prices one unit closing one deficit. Lowering the private cost of reaching the score can recruit additional below-threshold units into gaming, so conserved per-gamer harm does not imply conserved population harm.

Declared harm rates. The h_j are contract inputs, hidden from the scorecard but declared or estimated by the analyst. Without them there is no diagnostic — only the affordability statement.

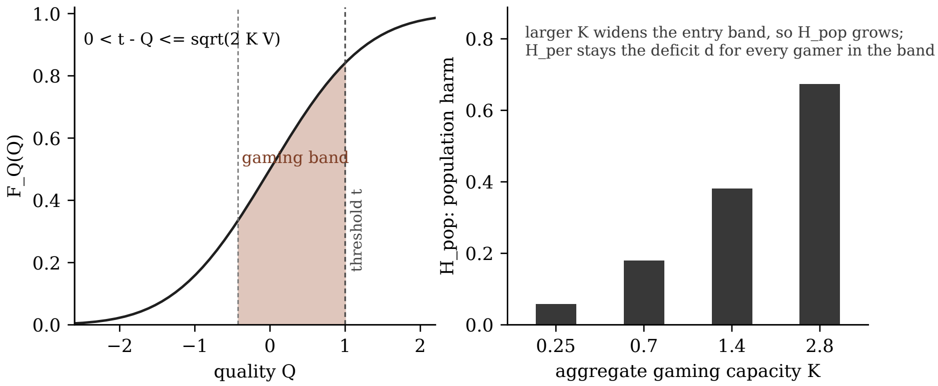

Figure 3: The gaming band separates fixed-deficit per-gamer harm H_{per} from population harm H_{pop}. More capacity widens entry without by itself changing the per-gamer exchange-rate formula.

The mechanics deferred here — population entry, conjunctive rules, and the step from per-agent harm to welfare — are taken up at the end of the next part.

5 Supporting Calculations

The remaining closed results support the exchange-rate diagnostic; none is an independent headline. The selection bounds say when an intervention reading is licensed at all — score movement with no unit acting is a different mechanism. The affordability results say which channels a design makes cheap, the input the diagnostic prices. The hardening result treats capacity reduction as measurement design in time. The response-shape taxonomy and the aggregation-and-entry mechanics mark where the fixed-deficit frame ends.

5.1 Selection: the channel to exclude first

The same score path the diagnostic explains can be produced with no unit acting at all: pure selection reweights a fixed population. Before any intervention calculation is licensed, the contract has to rule selection out or bound it. Selection is the regime where policy changes only the weights over a fixed baseline, so baseline variance and reweighting intensity are enough to bound hidden drift.

Before the theorem can be used, the contract must supply a baseline law, a pure selection channel, hidden coordinates or a value metric, and enough integrable moments. Pure selection means $K_\theta = K_0$ and the policy changes only the weights, with the induced law absolutely continuous with respect to baseline, $\mu_\theta \ll \mu_0$. Write $L = d\frac{\mu_\theta}{d}\mu_0$ and $\delta = \|L - 1\|_{L^2(\mu_0)}$, and call $B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu_0}[H]$ the selection drift of the hidden vector H.

T1. Coordinate-explicit selection bound. For hidden coordinates H_i with finite second moments and baseline standard deviations s_i,

$$\left|\mathbb{E}_{\mu_\theta}[H_i] - \mathbb{E}_{\mu_0}[H_i]\right| \le \delta s_i$$

for each coordinate. After a Euclidean coordinate metric is declared,

$$\left\|B_{H(\theta)}\right\|_2 \le \delta \|s\|_2$$

.

T2. Value-weighted/operator selection bound. For a declared scalar value direction v and covariance Σ_H,

$$\left|v \cdot B_{H(\theta)}\right| \le \delta \sqrt{v^T \Sigma_H v}$$

. For a declared norm,

$$\left\|B_{H(\theta)}\right\|_V \le \delta \sup_{\|v\|_{V,*} \le 1} \sqrt{v^T \Sigma_H v}$$

.

These are Hilbert-space Cauchy–Schwarz bounds because hidden drift is an inner product: it pairs the reweighting residual $L-1$ with the centered hidden variable, and Cauchy–Schwarz turns the pairing into an envelope. The bounds give drift envelopes for pure reweighting. They do not identify the hidden coordinates, the value weights, or the welfare object. They do not apply to fixed-type response changes.

Covariance enters as a local velocity and nothing more. Along a valid exponential tilt — one whose $\exp(\beta P)$ stays normalizable over the pressure range — the derivative at zero pressure is a covariance. At finite pressure the whole tilted path, tail shape, and moment-generating domain matter. $H = Z^2 - 1$ is the zero-covariance counterexample: baseline covariance zero, finite-pressure drift nonzero.

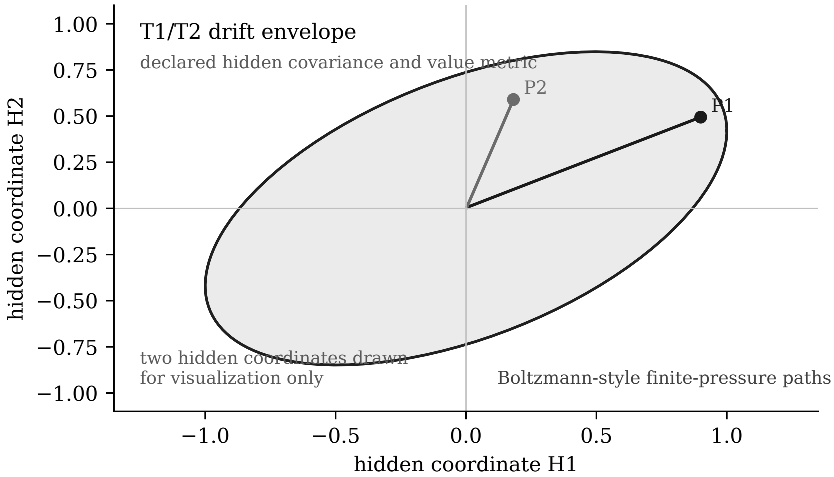

Figure 4: T1/T2 give a declared hidden-space envelope. Boltzmann-style pressure paths are trajectories inside that declared geometry, not replacements for the finite-pressure bound.

5.2 Intervention: when a gaming channel activates

Once fixed types can act, baseline-distribution bounds are no longer the right object; the relevant primitive is action affordability under costs and stakes. The affordability results say which channels a design makes cheap enough to activate — the input the exchange-rate diagnostic consumes.

Before an intervention calculation can be used, the contract must supply a fixed-type action model: actions a, private cost $c(a)$, proxy gain $w \cdot a$, score deficit d, and stakes V. The calculation measures the affordability of proxy movement under that model.

> **T3. Quadratic Stackelberg wedge.** In the one-dimensional noiseless threshold toy with quality Q, action $a \geq 0$, private cost $\frac{a^2}{2\kappa}$, threshold t, and stakes V, gaming by a below-threshold unit is privately worthwhile exactly when $t - Q \leq \sqrt{2\kappa V}$.

In prose: a below-threshold unit must buy the deficit $d = t - Q$; the cheapest passing action costs $\frac{d^2}{2\kappa}$; gaming is worthwhile exactly when that cost is at most V. The wedge $\sqrt{2\kappa V}$ is a signature of this quadratic toy. It is not a universal intervention law and should not be read as an RLHF, benchmark, or organizational theorem unless the missing primitives — action, cost, stakes, and pass condition — have been declared.

> **T4. Convex score-deficit budget.** With finite-dimensional action space, closed proper convex cost c, linear proxy gain $w \cdot a$, and regularity for convex duality, define $m(d) = \inf_a \{c(a) : w \cdot a \geq d\}$. Then
>
> $$m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)]$$
>
> . Gaming under stakes V is feasible exactly when $m(d) \leq V$ in this declared private-cost model.

In T4, c^* is the convex conjugate of the cost and λ is the multiplier pricing the score-deficit constraint. The budget licenses private affordability under the declared model and nothing more; the boundary between affordability and hidden harm is drawn with the exchange-rate diagnostic.

5.3 Adaptive hardening: measurement design in time

Hardening is the dynamic face of the design question: instead of changing which channels are measured, the designer reduces channel capacities over time and asks when gaming stops being feasible. The result is narrow by design. The contract is fixed finite measured set M, fixed deficit d, fixed stakes V, fixed weights, additive proxy gain, separable quadratic costs, deterministic observation, and monotone hardening of capacities $\kappa_{j,t}$.

> **T6. Deterministic adaptive-hardening capacity boundary.** Let
>
> $$S_{t(M)} = \sum_{j \in M} \kappa_{j,t} w_j^2$$
>
> and $T = \frac{d^2}{2V}$. At time t, fixed-deficit gaming is feasible exactly when $S_{t(M)} \geq T$. Hardening reaches no-gaming exactly when $S_{t(M)} < T$. A progress-aware largest-action multiplicative rule terminates in finite time when channels are finite, positive-weight, and floor capacity satisfies $S_{\text{floor}(M)} < T$.

Nothing broader is licensed. The theorem does not cover stochastic observation, arbitrary hardening rules, changing measured sets, changing deficits, changing stakes, nonconvex costs, shared bottlenecks, cycles, welfare comparison, or policy optimality. The immediate anti-transfer is noisy observation: once the update rule can chase noise, the deterministic threshold is no longer a stopping theorem.

5.4 Response shape, conditionally

The project did not prove a generic residual-shape law. What survives is a conditional taxonomy:

- Quadratic costs select the minimum-cost direction, often proportional to Cw in a smooth unconstrained model.
- Fixed charges create entry thresholds and active-set comparisons.
- Caps produce spillover only after a channel saturates.

- Low-rank affordances constrain response to an image, but do not choose a basis-invariant “simple” direction by themselves.
- Search-prior claims require a coding language or search process fixed before the failure is observed.

These are not theorem transfers between domains. They are ways to turn a response-shape conjecture into a declared model with falsifiers.

5.5 Aggregation and entry

Three mechanisms move harm outside the fixed-deficit frame of the exchange-rate diagnostic, and each needs its own declared rule.

Population entry. The diagnostic prices one unit closing one deficit. Lowering the private cost of reaching the score — adding cheap channels, raising capacities — can recruit additional below-threshold units into gaming. Per-gamer harm can be conserved while population harm grows on the entry margin; the gaming band of Figure 3 separates the two quantities.

Conjunctive aggregation. Requiring every measured component to clear a bar is a different aggregation rule with different comparative statics: harm can grow with the number of components because gates are cleared, not traded off. Nothing in the additive formula transfers without re-derivation.

From harm to welfare. Per-agent harm, population harm, and welfare are three different objects. Moving between them needs the declared value model — who counts, how harms aggregate, and what offsets are allowed. The contract treats that as a separate declaration, never a corollary.

The supporting calculations close here: they say when the exchange-rate frame applies, what feeds it, and where it ends.

6 Prior Work and Stress Tests

This part demonstrates the same claim twice: reduction, not unification. First on the literature — each nearby formalism supplies some fields of the response-modeling contract and omits others, so what transfers is a primitive, never a theorem. Then on cases — MMLU, hospital readmissions, and scientific metrics are run against the question the book is organized around: what would the exchange-rate audit need here, and what does the contract refuse without it?

6.1 Genealogy

Goodhart, Campbell, Strathern, and Manheim–Garrabrant are genealogy, not proof sources for the calculations above. Goodhart's original macroeconomic warning concerns policy-contaminated regularities [8]; the Lucas critique is its companion precedent, arguing that policy change invalidates the estimated relations policy relies on [14]. Campbell's warning concerns social indicators under decision pressure [3]. Strathern's formulation supplies the familiar compressed warning [23]. Manheim and Garrabrant provide a useful cause taxonomy [16].

This book does not replace that genealogy with a grander taxonomy. It uses the genealogy to ask which primitives are present: selection, response kernel, action cost, proxy/target separation, aggregation, hidden value, and evidence standard.

6.2 Formal analogues, primitive by primitive

The closest formal analogues supply some primitives and omit others. El-Mhamdi–Hoang and Majka–El-Mhamdi are scalar tail-conditioned selection anchors [6, 15]. Hardt et al. strategic classification supplies costly feature-change intervention [9]. Perdomo et al. performative prediction supplies deployment-induced response kernels [18]. Skalse et al. supply proxy/target and RL-specific response-kernel tools [13, 21]. Holmstrom–Milgrom supplies the closest multitask incentive analogue for aggregation and action costs [12]. Cawley–Talbot, reusable holdout, leaderboard work, and optimizer's curse supply selection and evidence standards [2, 4, 5, 19, 20, 22]. Pan and Gao provide empirical response-geometry candidates, not plug-in parameters [7, 17].

6.3 What each discipline contributes vs. omits

Source family	Primitives supplied	Primitives omitted	Licensed transfer
Tail-conditioned Goodhart	Selection, scalar proxy/target, tail assumptions.	Vector aggregation, intervention, hidden welfare.	Sharp scalar asymptotics where the tail contract holds; otherwise only a reweighting analogy.

Source family	Primitives supplied	Primitives omitted	Licensed transfer
Strategic classification and performative prediction	Response kernels, costly action or endogenous deployment.	General hidden value, scorecard exchange rates, marginal identification.	Intervention framing after action/ cost or deployment response is declared.
RL reward hacking	Proxy/target separation, MDP occupancy geometry, optimization pressure.	Non-MDP domains, empirical welfare, generic cost parameter.	RL-specific response calculations only inside the MDP/occupancy contract.
Multitask incentives	Aggregation and multidimensional effort/cost.	The book's value-metric contract, non-LEN settings unless separately shown.	External precedent for the scorecard exchange-rate question, not a proof of this book's theorem.
Adaptive holdout and leaderboard hygiene	Selection/evidence standards for repeated benchmark use.	Fixed-type fine-tuning, contamination, welfare, action cost.	Evidence discipline for selection over fixed candidates and public/private score interpretation.
Reward overoptimization empirics	Candidate search and optimization-pressure variables.	A defended mapping from model size, KL, data access, or prize to κ or V.	Hypothesis generators for response geometry; no automatic theorem transfer.
Empirical exchange-rate estimates (placeholder)	Candidate evidence for per-channel harm rates h_j: hospital readmissions versus mortality, education audit-test transfer, public-target gaming, reward-model overoptimization curves.	Identified per-channel rates; policy-variation designs that separate channel loads from harm movement.	None yet. This row family is a placeholder for a parallel empirical track; the diagnostic's measured content lands here.

6.4 Reduction, not unification

No single equation unifies these literatures. The framework's claim is narrower: each formalism fills different fields of the response-modeling contract. The contract says which primitive is being borrowed and which transfer fails. A source about adaptive holdouts does not identify hidden welfare. A source about costly feature change does not identify value weights. A source about multitask contracts does not make "more metrics" good or bad in a new application.

The same reduction discipline now runs on three concrete cases. Each case asks what the exchange-rate audit would need and what the contract refuses without it.

6.5 MMLU

MMLU is useful because it is not a single mechanism [10]. A public benchmark score can induce fixed-checkpoint model selection, repeated adaptive testing, prompt search, finetuning on benchmark-like data, contamination, reward-model overoptimization, or genuine capability transfer.

The contract separates these mechanisms. Fixed-checkpoint comparison is a selection problem: candidate checkpoints are types, the score is a proxy, and the hidden target is performance away from the benchmark. Finetuning, prompt search, contamination, or reward optimization is intervention/search only after an action/search geometry is declared. MMLU alone supplies neither κ nor stakes V, and it does not supply hidden welfare. T3/T4 apply only after a cost model and score deficit are declared. T5 applies only after benchmark components are modeled as additive channels with harm exchange rates.

The stress test is whether the evidence can distinguish fixed-candidate selection from post-exposure adaptation. Necessary observations include benchmark access, training data lineage, prompt/search budget, repeated-query history, transfer to non-MMLU tasks, and failure modes that move when the benchmark format changes.

6.6 Hospital readmissions

For a readmission scorecard, score improvement is compatible with hospitals leaving the comparison pool, coding changes, observation-status changes, delayed admissions, patient avoidance, better follow-up care, or mixtures.

What the exchange-rate audit would need here is concrete: effective score weights w, stakes V, response ease κ for coding, discharge timing, follow-up, and patient selection channels, hidden harm rates h, and the signal adequacy of the measured components. If these are unavailable, the design consequence is not "assume the score is bad." It is: do not use the toy diagnostic, collect action traces, monitor hidden patient outcomes where possible, pilot or lower leverage, and mark the missing primitive.

This is an evidence contract, not policy advice. The framework can say that aggregate readmission movement is insufficient to credit patient-welfare improvement. It cannot rank hospital policies without the clinical and value model the contract explicitly requires.

6.7 Scientific metrics

Publication counts, citations, grants, venue prestige, and rankings are already covered by responsible-metrics warnings such as DORA [1] and the Leiden Manifesto [11]. The framework does not claim novelty for the warning that crude metrics can distort science.

The stress test is discriminator value. A scientific scorecard application must separate selection over researchers, labs, fields, and institutions from fixed-researcher response. It must distinguish harmful proxy manufacture from harmless metadata/discoverability repair and from genuine improvements in methods, robustness, replication value, useful synthesis, or long-run uptake.

The contract refuses broad research-value inference from score movement. If an institution cannot declare the target, aggregation, action traces, and hidden value model before interpreting a score rise, the framework returns "no licensed hidden-value claim."

What the three cases share is the protocol shape: collect the discriminating traces, declare the primitives, and let the contract say which claim is licensed. The next part states that protocol in a form meant to be used.

7 Using the Framework

7.1 What to do before crediting score movement

Wentworth's practical warning is that experiments often measure something other than what the experimenter thinks they are measuring: an unexpected confounder, a secondary channel, a sampling artifact, or an operational detail that happens to track the named quantity [24]. The proposed repair is not to stare harder at the headline metric. It is to measure many auxiliary traces so the unexpected channel has somewhere to show up.

This book agrees with the first half and adds a second step. Auxiliary traces are necessary for discovery, but they are not sufficient for interpretation. They must feed a declared response contract. The combined rule is: collect the firehose for discovery, then use the contract for claim licensing. Without the firehose, the analyst misses the confounder. Without the contract, the analyst has many traces and still no disciplined statement of which score movement counts as selection, fixed-type response, proxy repair, harmful gaming, real improvement, or a mixture.

The practical workflow is therefore not to ask first, "did the score improve?" Ask instead: what changed, who changed, by what channel, at what cost, with what hidden target evidence, and what observation would make this interpretation fail? A score rise is an observation to be explained, not a conclusion to be credited.

Audit field	**Entry before crediting score movement**
Observed score movement	Magnitude, timing, affected units, score components, and whether the movement is marginal, thresholded, ranked, or aggregate.
Candidate mechanisms	Selection, fixed-type response, proxy repair, harmful gaming, real improvement, or an explicit mixture.
Required discriminator traces	Repeated-type evidence, exposure timing, action logs, search or query budget, component-level score changes, participation changes, and off-score outcomes that separate nearby mechanisms.
Hidden value/harm evidence	Declared target coordinates, value weights, harm rates, residual measures, or domain outcomes; if absent, mark the hidden-value claim unlicensed.
Missing primitives and operational consequence	The undeclared type space, baseline law, response channel, action cost, aggregation rule, or hidden value model, plus the resulting consequence: collect traces, lower leverage, pilot, or withhold the claim.
Licensed claim	The narrow statement supported by the declared contract: for example, pure-selection drift envelope, private affordability under a cost model, exchange-rate diagnostic, or no hidden-value conclusion.

Audit field	Entry before crediting score movement
Blocked claim	The stronger statement not supported: welfare improvement, target improvement, policy optimality, channel identification, or generic anti-metric advice.
Contract-failure condition	A concrete observation that would defeat the interpretation, such as action traces inconsistent with the named channel, hidden outcomes moving opposite the claimed target, or theorem conditions failing.

This sheet is smaller than the full application template. Its job is to slow down the common interpretive jump. If the observed movement is a readmission decline, a benchmark rise, a citation increase, or a ranking gain, the first audit question is the same: what response channel could have produced this movement? The second is which traces would discriminate that channel from its nearest rivals. The third is which hidden value or harm evidence is present, and which claim remains blocked if it is absent.

The conclusion can be positive, but it has to be narrow. "The score improved and auxiliary outcomes moved in the declared target direction under stable exposure" is a different claim from "the policy improved welfare." "The action logs fit a low-cost proxy-repair story" is a different claim from "the metric is safe." "The selection envelope bounds hidden coordinate drift under the stated law" is a different claim from "selection is harmless." Practical use of the framework is mostly this discipline of replacing a large conclusion with the smaller one the evidence actually licenses.

8 Refusals, Falsifiers, and the Open Agenda

8.1 Anti-applications

The framework is least useful when its primitives cannot be stably declared. Concrete anti-applications include:

- A rapidly shifting online platform where type space, measured set, policy exposure, and available actions all change between measurements.
- A black-box institutional ranking where effective weights and thresholds are unobservable and participants cannot tell which action channel is being rewarded.
- A high-stakes public decision where stakeholders demand a welfare verdict but no defensible hidden value model exists.

In these cases, the contract returns "no verdict." That is not a hedge; it is the framework refusing to convert score movement into causal mechanism or welfare language without the primitives that would make the claim testable.

8.2 Falsifiers

The contract is falsifiable as a modeling discipline, but the failure point has to be located honestly. The theorems themselves cannot be empirically violated while their hypotheses hold; they are proved. What can fail is the claim the contract makes about a domain: that independently audited declarations of channel, costs, stakes, and harm rates track the response mechanism well enough for the licensed calculation to predict response shape or direction. A concrete falsifier is therefore a domain where the primitives are declared and audited in advance, response channel and action traces are observable enough to distinguish nearby mechanisms, and the licensed calculation still systematically predicts wrongly while a simpler score-only rule predicts correctly.

Examples include:

- A setting independently audited as pure selection, with the declared moment and absolute-continuity conditions checked, where hidden drift repeatedly exceeds the δs_i envelope. The bound cannot fail under its hypotheses, so repeated excess drift would show that the audit and the contract's response-channel field failed to detect a fixed-type response channel.
- A hardening setting audited as having stable M, d, V, weights, separable quadratic costs, and deterministic observation, where gaming remains feasible after $S_{t(M)} < \frac{d^2}{2V}$ or stops while $S_{t(M)} \geq \frac{d^2}{2V}$ — showing the audited capacities or observation model were not the ones the agents faced.
- An additive scorecard with defended w_j, κ_j, and h_j where fixed-deficit per-agent harm is conserved across active measured sets even though the exchange-rate condition $h_j = cw_j$ fails, or moves even though the condition holds — showing the declared channels, costs, or harm rates were not the ones generating the response.

Such failures, if they survived audit, would not call for prose repair. They would mean the contract fields do not track the response mechanisms they were designed to track — that disciplined declaration does not buy predictive power. That is the falsifiable content of the framework.

8.3 The residual-shape conjecture

The signature open problem is the residual-shape conjecture: when does repeated proxy repair drive hidden failure toward a predictable residual shape, such as a low-complexity attractor? The current answer is negative unless a mechanism is named first.

Resolving it requires at least five declarations: a response mechanism, a complexity or shape functional fixed before inspection, a policy-update rule, a composition rule for repeated repair, and a failure condition. It would count as progress to prove a sparse, low-rank, or low-description-length attractor inside one of those contracts. It would not count to observe a simple-looking failure after the fact and relabel it as the attractor.

8.4 Composition, identification, and information

Three structural gaps remain.

First, channels compose. Real cases mix selection, fixed-type response, proxy repair, and real improvement. The current contract names mixtures but does not give a general calculus for them.

Second, primitives are not automatically identifiable. Marginal score movement does not identify W_θ versus K_θ, hidden harm, action costs, or aggregation. The framework needs identification toys and evidence thresholds for when a primitive can be treated as declared rather than guessed. The sharpest version of this gap belongs to the exchange-rate diagnostic: under what policy-variation designs are per-channel harm rates h_j identifiable from aggregate score and harm movement? Until that question has an answer, the diagnostic's empirical row stays a placeholder.

Third, the L^2 selection bound may not be the portable final form. An information-theoretic restatement could travel further if it preserves the distinction between coordinate-explicit drift, declared value metrics, and finite-pressure path behavior.

8.5 Toolkit gap

A practitioner should not have to re-derive the contract every time. The missing toolkit has three parts: a primitive-elicitation protocol, worked exchange-rate audits in real scorecards, and small identification examples that show what observations distinguish selection, intervention, proxy repair, and real improvement.

Until those tools exist, the framework is best read as a claim-license discipline and a theorem inventory. It tells a reader what would have to be true before a Goodhart calculation travels.

Appendix: The Killed-Claims Gallery

The full set of failed stronger claims behind the no-generic-law part. Each row records the refuted claim and what survives it; load-bearing rows appear in paired form in the main text.

Failed claim	**What survives**	**Tier**
Hidden harm scales with $\dim(\ker \varphi)$ by itself.	A coupling question: unmeasured dimensions matter through dependence on the selected proxy and the declared value metric. Independent hidden coordinates do not move under thresholding.	Counterexample
More measured dimensions means more (or less) hidden error.	Aggregation-conditional comparative statics: additive and conjunctive rules give opposite signs from the same action channels.	Counterexample
Signed aggregate hidden error measures damage.	A declared value functional, norm, tail risk, or domain loss; signed hidden movements can cancel.	Counterexample
Baseline covariance predicts finite-pressure response.	Covariance as the zero-pressure derivative. With $P = Z$ and $H = Z^2 - 1$, baseline covariance is zero while threshold or Boltzmann selection moves H.	Counterexample
Convex affordability is welfare.	Affordability claims under the declared private-cost model; welfare needs a declared harm functional.	Counterexample
Absolute continuity marks the intervention boundary.	Absolute continuity as a hypothesis of the reweighting theorem; fixed-type action changes can remain absolutely continuous with respect to the baseline law.	Boundary
The selection/intervention split is identifiable from marginal data.	The split as a declared field, relative to a defended type representation, supported by repeated-type, exposure, action-trace, or structural evidence.	Boundary
The selection bound is coordinate-free by itself.	A coordinate-explicit bound until a value norm or scalar value functional is named.	Boundary
Optimization pressure drives failure toward minimum-complexity shapes.	Conditional response-shape statements: selection follows baseline tails; intervention follows cost, search, caps, fixed charges, and affordances.	Boundary

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